become "past due"). Using Bayes' formula, he has structured the problem as follows:

$$P(\text{Event} \mid \text{Information}) = \frac{P(\text{Information} \mid \text{Event})}{P(\text{Information})}P(\text{Event}),$$

where the event (A) is "timely repayment" and the information (B) is having a "good credit report."

Bronson estimates that the unconditional probability of receiving timely payment, $P(A)$, is 0.90 and that the unconditional probability of having a good credit report, $P(B)$, is 0.80. The probability of having a good credit report given that borrowers paid on time, $P(B \mid A)$, is 0.85.

What is the probability that applicants with good credit reports will repay in a timely manner?

A. 0.720

B. 0.944

C. 0.956

Solution:

The correct answer is C. The probability of timely repayment given a good credit report, $P(A \mid B)$, is

$$P(A|B) = \frac{P(B|A)}{P(B)}P(A) = \frac{0.85}{0.80} \times 0.90 = 0.956$$

2. You have developed a set of criteria for evaluating distressed credits. Companies that do not receive a passing score are classed as likely to go bankrupt within 12 months. You gathered the following information when validating the criteria:

- Forty percent of the companies to which the test is administered will go bankrupt within 12 months: $P(\textit{non-survivor}) = 0.40$.
- Fifty-five percent of the companies to which the test is administered pass it: $P(\textit{pass test}) = 0.55$.
- The probability that a company will pass the test given that it will subsequently survive 12 months, is 0.85: $P(\textit{pass test} \mid \textit{survivor}) = 0.85$.

Using the information validating your criteria, calculate the following:

A. What is P(*pass test* | *non-survivor*)?

B. Using Bayes' formula, calculate the probability that a company is a survivor, given that it passes the test; that is, calculate $P(\textit{survivor} \mid \textit{pass test})$.

C. What is the probability that a company is a *non-survivor*, given that it fails the test?

D. Is the test effective?

Solution:

A. We can set up the equation using the total probability rule:

$$P(\textit{pass test}) = P(\textit{pass test}|\textit{survivor})P(\textit{survivor})$$